BharatBazaar AI

AI-Powered Market Intelligence for 15 Million+ Indian Kirana Stores

23	13	8	6	10
Features	APIs	AI Features	Languages	Cities

GitHub (Web): github.com/m-zest/bharat_bazaar_v2.4.8 *GitHub (Mobile): github.com/ParityAI/bharat_bazaar_mobile*

Team ParityAI | ***Mohammad Zeeshan** — AI Researcher* | ***Afzal Hussain** — Software Developer*

01 The Problem

The people who feed India have zero data intelligence.

Pricing by Gut Feeling

~50K lost/yr

Store owners set prices based on instinct, losing potential revenue every single day to competitors who know the market.

Paper Notebooks (Bahi-Khata)

0% visibility

Inventory tracked in handwritten bahi-khata. No visibility into what sells, what rots, or what to reorder.

English-Only Tools

70% excluded

70% of store owners prefer regional languages. Enterprise tools cost 50K+/month and speak only English.

No Digital Presence

500M on WhatsApp

500 million Indians on WhatsApp. Yet most stores have zero digital ordering, zero online price discovery.

15M+ Kirana Stores | $1.3 Trillion Market | 4th Largest Retail Market Globally | 90% Undigitized

02 Our Solution — Three Platforms

BharatBazaar AI answers 3 questions every store owner asks — across Web, Mobile, and WhatsApp.

"What is in my store?"

Khata Bill Scanner

Photo of bill > AI reads Hindi + English > extracts items, quantities, prices > one tap adds to DynamoDB inventory.

"What is selling?"

Live Dashboard + GST Invoices

Every invoice auto-records the sale. Dashboard shows revenue, top sellers, trends, AI insights, weather, festivals in real-time.

"What should I price?"

AI Pricing Engine

AI analyzes cost + city + competitors + weather + festivals. Returns 3 strategies: Competitive, Balanced, Premium with confidence.

Key Insight: Store owner's daily routine (scanning bills, generating invoices, chatting on WhatsApp) IS the data input. Zero spreadsheets. Zero data entry.

Available on: **Web App** + **Mobile App** + **WhatsApp Bot (Twilio)** | Login: admin / admin

03 Your Daily Actions = Your Data

No data entry. No spreadsheets. The store owner's daily routine automatically builds the intelligence layer.

1

Setup Your Store

Catalog personalized

Tell us what you sell during onboarding. Catalog personalized with real products, prices, and sell rates.

2

Scan Purchase Bills

Stock auto-updated

Photograph any wholesale bill. AI extracts products, quantities, prices. Inventory grows automatically.

3

Generate Sales Invoices

Sales tracked live

Every GST invoice = a sale recorded. Revenue, items sold, top sellers — all updated in real-time.

4

Chat on WhatsApp

Orders processed

"Order 50 Surf Excel" or "Price for Tata Salt" — AI processes every message into structured data.

No manual data entry. Every interaction becomes business intelligence — stored in DynamoDB, analyzed by Bedrock AI.

Store Setup > Stock In (Bills + Orders) > Sales Out (Invoices) > AI Intelligence (Pricing + Forecasts + Alerts) > Cycle Repeats

04 Why AI is Required

Each feature needs AI — not rules, not spreadsheets, not if-then logic.

Dynamic Pricing

5+ variables: cost, city tier, season, festival, competitors. No static formula works across India's diverse markets.

26% margin improvement

Multilingual Content

Descriptions need cultural adaptation, not translation. Tamil messaging differs from Hindi — not just in words, but in tone and context.

6 native languages

Hinglish Sentiment

"Packaging tuti hui thi, but product accha hai" — rule-based NLP fails completely on code-mixed Indian text.

Mixed-language accuracy

Bill Scanning (Vision)

Handwritten Hindi bills, varying formats, Hindi numerals. OCR alone cannot understand context or Indian product names.

Zero data entry

Business Advisor Chat

"Baarish ho rahi hai, kya mangau?" needs reasoning about weather patterns, demand correlation, and current inventory.

Contextual Hinglish advice

Voice Input / Output

Target users have low text literacy. Voice in Hindi, Tamil, Telugu is their natural mode. AI transcribes and responds aloud.

8 languages supported

05 The Wow Factors

Features that prove we understand the actual Indian kirana store owner

VOICE

Hindi Voice I/O

Shopkeeper speaks: "Munim-ji, daal ka rate batao"

AI transcribes Hindi, processes, responds in Hindi voice

8 languages: Hindi, Tamil, Telugu, Bengali, Marathi + more

Web Speech API — zero backend cost, works on any phone

VISION AI

Khata Bill Scanner

Snap wholesale bill photo with phone camera

Gemini 1.5 Flash Vision reads Hindi + English text

Extracts names, quantities, prices — even Hindi numerals

One tap updates DynamoDB with source badge tracking

WHATSAPP

WhatsApp Bot

Judge texts from their OWN phone — live demo

"kitna Parle-G bacha hai?" — bot checks DynamoDB

Replies: "5 left. Reorder?" with rich formatting

500M Indians on WhatsApp. This IS the product.

06 AWS Architecture — Deployed

FRONTEND

React 18 + TypeScript + Vite + Tailwind CSS + Framer Motion + Recharts

Web App (23 pages), Mobile PWA, WhatsApp Bot, Voice I/O

COMPUTE

AWS CloudFront (CDN) + S3 Static Hosting + Docker (App Runner ready)

Both web and mobile deployed on AWS CloudFront — d3j4u51h5o0dhm.cloudfront.net

API LAYER

Express.js — 13 REST Endpoints + CORS + Rate Limiting

11 handlers, all Lambda-typed with @types/aws-lambda

AI ENGINE

Amazon Bedrock (Claude 3 Haiku + Nova Lite) + Gemini 1.5 Flash

4-tier fallback chain — user never sees an error

DATABASE

Amazon DynamoDB — Single-Table Design (PK/SK) + PAY_PER_REQUEST + GSI

Auto-creates table, multi-tenant ready, scales to millions

AWS SERVICES

- Bedrock — AI inference
- DynamoDB — NoSQL DB
- Bedrock Vision — Bill OCR
- CloudFront — CDN (deployed)
- S3 — Static hosting
- IAM — Security
- CloudWatch — Monitoring
- Gemini — AI fallback
- App Runner — Docker ready

4-TIER FALLBACK: Claude 3 Haiku (ap-south-1) > Nova Lite (us-east-1) > Gemini 1.5 Flash > Smart Demo Mode | if(model.fails) > auto_retry(next) > user sees zero errors

07 WhatsApp-First Platform

500M+ Indians on WhatsApp. We built every feature to work inside the chat. No downloads, no learning curve.

Already on every phone
No downloads, no signups needed

Voice messages supported
Send voice note in Hindi, AI understands

Works from shop counter
Quick replies between customers

Works on 2G/3G
Text-based AI, even on slow networks

EVERY FEATURE ON WHATSAPP

Inventory Check "stock"
Real-time stock levels

Smart Pricing "price [item]"
AI price suggestions

Place Orders "order [items]"
Restock from wholesalers

Scan Bills Send photo
OCR via camera

Daily Reports "report"
Sales + profit summary

GST Invoice "bill [customer]"
Generate + share

Customer Khata "khata [name]"
Track credit + payments

Multi-language "hindi / tamil"
Switch anytime

500M+
WhatsApp users

15M+
Kirana stores

0
App downloads

2G
Works on slow nets

The web app is the demo. The mobile app is the vision. WhatsApp IS the product.

08 India-First Intelligence

6 Native Languages

Hindi, Tamil, Bengali, Gujarati, Marathi, English — culturally adapted, not just translated. Content Studio generates for 6 platforms.

10 City Profiles with PPP

Mumbai (High) to Lucknow (Medium) — same product gets fundamentally different AI recommendation per city and purchasing power.

Festival-Aware Demand

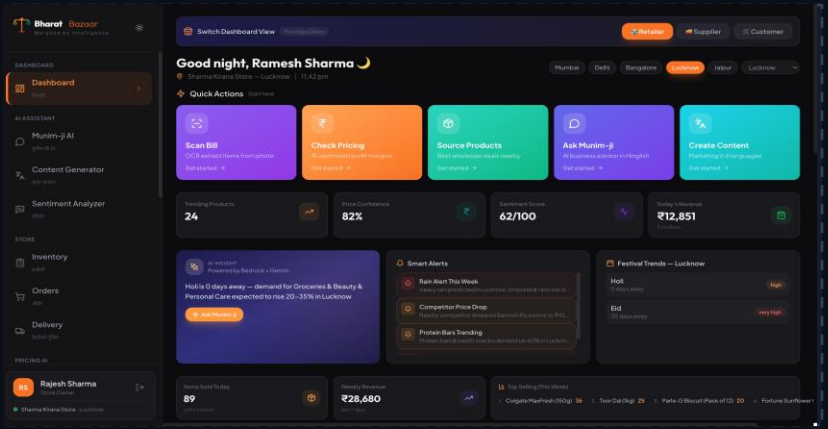
Diwali, Holi, Eid, Pongal, Navratri, Christmas — demand multipliers and impact scores per region. Dashboard shows upcoming alerts.

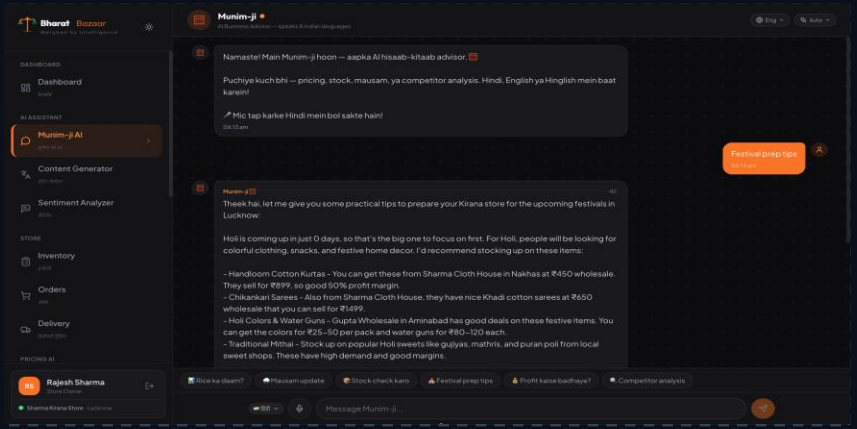
Weather-Driven Insights

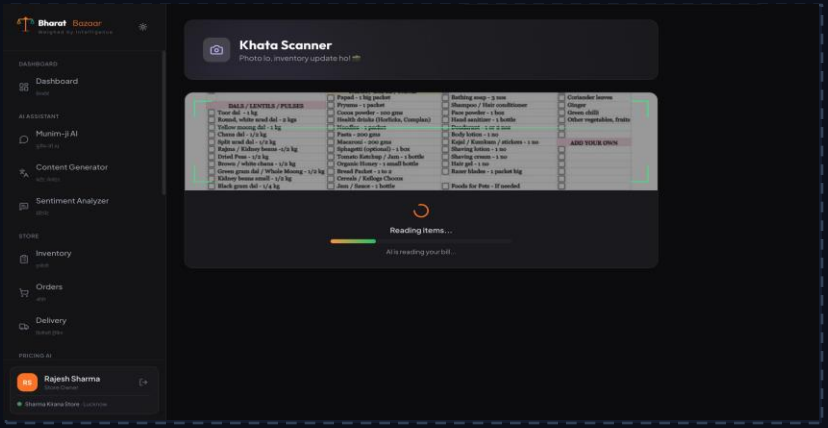
Rain forecast = +15% dal demand. Hot weather = +20% beverages. AI factors this into every pricing recommendation automatically.

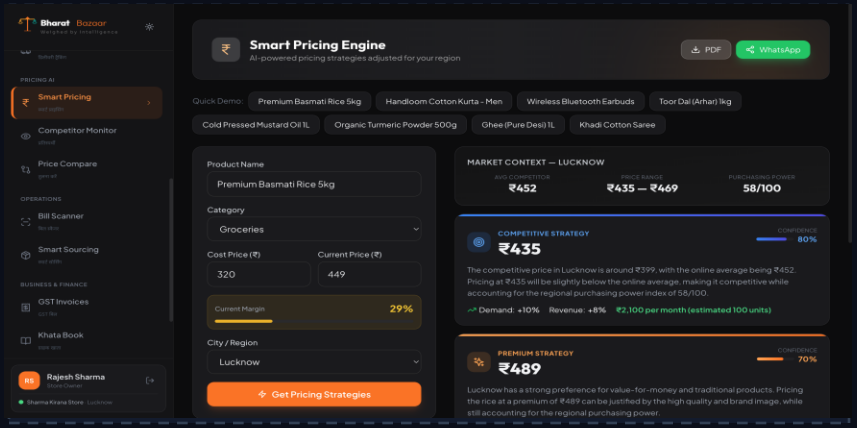
We did not translate English tools into Hindi. We built an Indian tool from scratch.

09 Live Demo — Screenshots









10 One Platform. Three Interfaces.

We built a prototype for every surface Indian merchants use — web dashboards, mobile apps, and WhatsApp chatbots.

Web App

Full Dashboard Experience

- 23 interactive pages
- Real-time analytics
- GST invoice generation
- Dark/light theme

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Mobile App

Native-Feel PWA

- Camera bill scanning
- Voice commands
- Responsive design
- 5 key screens

https://d2a5rnm0qdxhtx.cloudfront.net/

WhatsApp Bot

Munim-ji AI

- Zero downloads needed
- Voice note support
- 6 languages
- Works on 2G

WhatsApp: Twilio Sandbox

Start on WhatsApp. Scale to Web and Mobile. All your data syncs across every platform.

11 Impact & Business Model

15M+	$1.3T	90%	26%
Kirana Stores	Market Size	Undigitized	Margin Boost

Free for Every Store. Start free. Grow when ready.

Every kirana store deserves AI intelligence — regardless of size or budget.

Free Forever

Current Plan

- Full platform access
- AI pricing suggestions
- Inventory management
- Invoice generation
- WhatsApp integration
- 6 languages

Pro Subscription

As business scales

- Priority AI models
- Advanced analytics
- Bulk sourcing marketplace
- API access + integrations
- Dedicated support
- Custom AI training

BharatBazaar AI

"Amazon has data science teams.
Kirana stores have BharatBazaar AI."

Try it Live

Web App: d3j4u51h5o0dhm.cloudfront.net Mobile App: https://d2a5rnm0qdxhtx.cloudfront.net/

Built on AWS: Bedrock | DynamoDB | CloudFront | S3 | IAM | CloudWatch | + Gemini 1.5 Flash | Docker | React 18 | TypeScript | Tailwind CSS